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CUDA:OpenCL translation

Invited Tal Kadosh Ayon Basumallik Niranjan Hasabnis Tomer Bitan
Armand Mousavi Le Chen Matthew Freihofer Sanjna Ravichandar Inbal Cohen
Bolet, Gregory Samyak Jhaveri Erel Kaplan Avraham Evdaev Yosef Goren
Soundararajan, Prema le.chen@anl.gov Gal Oren Roee Bar-Yadin ~~Peter Morales~~
~~Octavian Udrea~~ ~~Laura Titolo~~

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Summary

Augmentation and LLM evaluation harness design progressed with baseline results showing poor OpenMP pass rates while prioritizing anonymous submission requirements for the paper.

LLM Evaluation and Kernels

LLM evaluation harness design progressed with 5 types of augmentations, but aggressive augmentations L3 and L4 are currently failing when integrated. Baseline evaluation of 60 kernels revealed a dismal pass rate for OpenMP and an initial cross-API pilot showed poor results attributed to a lack of an AMD GPU for testing.

Paper Submission Requirements Finalized

The team decided to omit the exceptionally long kernel build times from the final report to focus on more relevant metrics. A new anonymous GitHub username, Powerbench, must be created for the paper's anonymous submission, and the paper must clearly justify the selection of kernels and APIs (Rodinia and Hecbench) as a representative subset of a larger survey.

Metrics and Model Selection

The final LLM evaluation strategy will focus on non-reasoning models and include both kernel execution time and overall program time with memory transfer metrics. The primary models selected for evaluation are GPT-4.1 and Llama 70B, with one more general-purpose model to be chosen from a coding leaderboard.

Details

- **Augmentation Status and LLM Evaluation Harness Design:** Samyak Jhaveri provided an update on the five types of augmentations, L0 to L4, which are designed to be increasingly aggressive to obfuscate the code, thereby forcing the LLM to rely on reasoning rather than memory for parallel code generation. Errol is assisting with the augmentation process, and Samyak Jhaveri has been designing the LLM evaluation harness, despite initial issues with the augmentation code being overly focused on OpenMP and causing errors for CUDA and other APIs ([00:00:00](#)). The current testing, using a 4070 GPU with 12 GB VRAM, includes 240 total CUDA to OpenMP and OpenCL pairs for translation evaluation ([00:01:40](#)).
- **Kernel Distribution and Baseline Evaluation Results:** The team is working with a distribution of kernels spanning physics, machine learning, image, graph, crypto, and sorting. In the baseline evaluation using Hecbench, 60 kernels have been tested so far, showing a pass rate for CUDA but not for OpenMP ([00:01:40](#)). The cross-API pilot using 20 Rodinia kernels resulted in a "dismal pass rate," attributed to the lack of an AMD GPU for testing ([00:03:03](#)).
- **Consideration of Build Time Data:** The individual build times for all kernels are exceptionally long, prompting discussion on whether this data should eventually be omitted from the report. Samyak Jhaveri agreed with Gal Oren's suggestion to consider emitting the exceptionally long build times. Gal Oren suggested that Errol should discuss the augmentation, but Samyak Jhaveri decided to continue with the parts they knew for sure first ([00:03:03](#)).
- **Anonymous Submission Requirements for the Paper:** Gal Oren instructed Samyak Jhaveri to create a new, anonymous username named Powerbench on GitHub and ensure all related material is under powerbench.github.io for the

paper's eventual anonymous submission. This website should also demonstrate the initial systematic steps taken, showcasing the pool of open-source codes and the mapping of which codes connect to which ([00:04:23](#)).

- **Justification for Kernel and API Selection:** Gal Oren emphasized the need to explain the selection process of kernels and APIs (Hecbench and Rodinia) in the paper and on the website to address potential reviewer questions ([00:05:52](#)). The claim should be that a larger survey of benchmarks was conducted, internal connections were mapped, and the current benchmark is a representative subset of that survey, allowing others to add their own kernels later ([00:07:40](#)).
- **Augmentation Integration and Bug Fixing:** Samyak Jhaveri reported that augmentations L3 and L4 are failing miserably when integrated with the evaluation harness, but they have identified the bugs and Errol is actively working on the fixes ([00:09:07](#)). Errol Kaplan confirmed that the initial code, intended for OpenMP, broke due to language-related symbols and reserved words in CUDA and OpenCL, but they expect the fixes to be completed by the following day ([00:10:17](#)).
- **Trade-offs in Augmentation Aggressiveness:** Erel Kaplan explained that a conservative approach to augmentation is necessary to ensure the code works across all given inputs, acknowledging that some variable names might not be changed ([00:10:17](#)). Gal Oren agreed with this conservative approach, noting that the paper will need to state the scope of the augmentation and potentially suggest that more aggressive augmentation could be part of future work ([00:11:42](#)).
- **Baseline Paper for Comparison:** Gal Oren identified the Paraval paper as the baseline for their work, instructing Samyak Jhaveri to read it but also focus on how their current work (Powerbench) represents a completely different evolution from Paraval ([00:12:55](#)). Gal Oren noted that Paraval uses a naive approach, is not augmented, contains very few kernels, and lacks a good explanation for the selected APIs and kernels, providing a point of critique for their paper ([00:14:06](#)).
- **Measuring Overall Program Time:** Samyak Jhaveri proposed reporting both kernel times and overall program time because LLM translation must be evaluated on its ability to handle data management, which is a major latency-introducing aspect of parallel code on GPUs ([00:15:05](#)). Gal Oren and Niranjan Hasabnis agreed to include both metrics, clarifying that the relevant end-to-end time for a GPU kernel involves kernel execution time plus memory

transfer time between the host and device ([00:16:18](#)). Samyak Jhaveri confirmed they will add the memory transfer time metrics from the Paracortex paper ([00:17:21](#)).

- **Model Selection for Translation Evaluation:** Gal Oren asked the group which models, besides GPT-4.1, should be included for evaluation, given the deadline. Samyak Jhaveri suggested including at least one open-source model, such as Llama ([00:18:20](#)). Tomer Bitan suggested an agentic model like Codex, but Gal Oren decided against agents and reasoning models for this paper to focus on the LLM's inherent knowledge and ability to handle augmented code, making Llama 70B and GPT-4.1 primary candidates ([00:19:31](#)).
- **Model Selection Strategy and Reasoning Feature:** The team agreed that selecting too many models should be avoided. Le Chen proposed checking coding capability leaderboards and selecting the top general-purpose models, such as Mistral and Qwen, to justify the selection in the paper ([00:21:41](#)). The discussion shifted to whether to include reasoning models or features, and Le Chen suggested starting without reasoning for all models for a fair comparison, which Samyak Jhaveri accepted ([00:24:55](#)).
- **Paper Outline and Benchmark Selection:** Samyak Jhaveri asked if they should attempt a third benchmark suite beyond Rodinia and Hecbench. Gal Oren reiterated that the paper must begin by systematically explaining the curation process: identifying all relevant open-source benchmarks, mapping relevant APIs, and justifying the use of a representative subset (Rodinia and Hecbench kernels) ([00:26:54](#)). They determined that they would take inspiration from Gal Oren's initial graph and systematic review, positioning the work as a necessary benchmark for parallel API translation, which is currently lacking ([00:28:10](#)) ([00:30:09](#)).
- **Emphasis on Related Work and Future Benchmarks:** The team agreed that the paper must include a detailed literature review section explaining how this work differs from prior research, particularly Paraval, which they view as more of an evaluation metric paper than a comprehensive benchmark paper ([00:32:26](#)). Samyak Jhaveri agreed to add ExaBench as a strong candidate for a third benchmark suite because it was a main kernel in Paraval, allowing for a comparison to previous work. They decided to focus first on creating an outline to identify gaps and areas for improvement rather than immediately adding more benchmarks ([00:34:39](#)).

- **Final Plan for Model Selection and Augmentation Justification:** The final strategy for LLM evaluation is to use non-reasoning models to test the LLM's inherent knowledge, justifying augmentation as a means to prevent reliance on memory. The selected models are GPT-4.1 and Llama 70B (with external hosting help), along with one more practical, general-purpose model from a leaderboard ([00:37:05](#)).
- **Update on Guidance Model Data Set Generation:** Tomer Bitan provided an update on the separate project of building a data set to train a guidance model (PRM) to assess the quality of reasoning steps based on latent vectors. The infrastructure is working, but generating the data set by running 220 Hecbench kernels through Llama 70B is slow, with each kernel taking 10 to 20 minutes ([00:39:25](#)).
- **Accelerating Data Set Generation:** Tomer Bitan is trying to supplement the code data set with an easier-to-make math-based data set (GSM8K extension) to pre-train the PRM, as they are concerned the Hecbench set will take too long and might not have enough kernels ([00:41:20](#)). Niranjana Hasabnis offered to check on providing one more parallel server from their cloud to accelerate the work ([00:43:27](#)).
- **Drafting the Paper Outline for Feedback:** Gal Oren instructed Tomer Bitan to immediately open a new Overleaf project and start writing the title, abstract, introduction, research questions, and methodology, even without final results, to help the group provide timely feedback and guidance for the risky project ([00:45:21](#)). This proactive writing will help them decide in two weeks whether to pivot or drop certain aspects of the plan ([00:47:20](#)).
- **Model Size Considerations for PRM Training:** Niranjana Hasabnis suggested using a smaller model, such as an 8 billion model, with a subset of Hecbench for initial testing to see if the idea works before scaling up to the 70 billion model ([00:48:09](#)). Tomer Bitan expressed concern that the 8 billion model performs poorly on Hecbench (producing incomplete code snippets), suggesting instead a gradual downgrade from 70 billion to 30 billion to find a model that still produces quality code ([00:49:11](#)).
- **ACL Paper Backup Plan:** Gal Oren reminded Erel Kaplan about the ACL paper acceptance notification on April 4th, which coincides with the Supercomputing submission deadline, instructing them to have the Supercomputing template ready to immediately move the work there if the ACL paper is not accepted

([00:51:56](#)). Niranjan Hasabnis offered to meet separately with Tomer Bitan to discuss the experimental process and findings in more detail ([00:53:09](#)).

- **Focusing on the Model's Applicability:** Niranjan Hasabnis raised concern about whether a reasoning technique that works on mathematical problems would translate effectively to parallel programming. Tomer Bitan found it difficult to abandon the parallel programming context, but acknowledged the suggestion to focus on the technique's general applicability across any domain first. They concluded that if the technique is a general one, a paper could be written about it, followed by a specific application to parallel programming ([00:58:17](#)).
- **Prioritizing Validation on Simpler Benchmarks:** Tomer Bitan agreed to start by attempting to quickly obtain results on the mathematical benchmark to validate the core idea, while not completely giving up on the parallel code application. Niranjan Hasabnis stressed the need to confirm that the idea itself works, noting that the current confidence is primarily based on intuition and extrapolation from other papers, without a full test demonstrating its effectiveness. They agreed it is worthwhile to check if the idea works on a simpler benchmark, such as mathematical reasoning, before tackling the more complex problem of parallel programming reasoning ([00:59:12](#)).
- **Next Steps for Experimentation and Review:** If the idea does not work on the math domain, there is a lower probability it will work on parallel programming, making the math evaluation a necessary first step. Tomer Bitan expressed that this clarification was helpful for organizing their thoughts. Niranjan Hasabnis offered to set up another meeting for a detailed discussion next week, requesting that Tomer Bitan share any experimental progress, even if full results are not yet available, to provide context before the next discussion ([01:00:06](#)).

Suggested next steps

- ☐ Erel Kaplan will fix the bugs with the L3 and L4 augmentations and work on making sure the code augments everything well without breaking anything, which should be completed by tomorrow.
- ☐ Samyak Jhaveri will open a new username named Powerbench in GitHub and make sure that everything is at that website for anonymous submission.
- ☐ Samyak Jhaveri will add a screen or tab to the Powerbench website demonstrating the initial step of taking all opensource codes and showcasing

which one connects to which one to explain the pool from which kernels are taken.

- ☐ Samyak Jhaveri will report overall program time in addition to kernel times, and will add the metrics from the Paracordex paper to the evaluation.
- ☐ Niranjan Hasabnis will come up with a way to measure kernel plus the host time for a GPU kernel.
- ☐ Samyak Jhaveri will create an outline of the paper and ensure it includes world-building, the curation rationale for selected benchmarks/APIs, and emphasizes the related work section.
- ☐ Niranjan Hasabnis will check with Alex from the code medicine on the provisioning part to see if Tomer Bitan can have another parallel GPU instance in mono.
- ☐ Tomer Bitan will make a small modification to the code to utilize multiple GPU instances for the math data set.
- ☐ Tomer Bitan will open a new overleaf project and begin writing the paper, including a title, fake abstract, intro, research question, and methodology.
- ☐ Niranjan Hasabnis will communicate via email with Tomer Bitan to understand his findings so far before discussing the next step for experimentation.
- ☐ Samyak Jhaveri will read the Paraval paper and keep in mind how the current work defers from it.
- ☐ Niranjan Hasabnis will set up a meeting with Tomer Bitan next week to discuss the experiments and understand the situation better.
- ☐ Tomer Bitan will evaluate the idea on a simpler benchmark, such as math reasoning, to verify if the idea works before trying to translate the results over to parallel codes.
- ☐ Tomer Bitan will share the results of experiments done so far with Niranjan Hasabnis.

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